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Smart Driver Assistance: Real-Time Drowsiness Detection Leveraging Facial Cues with MediaPipe and OpenCV

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Abstract

The primary aim of this research is to identify driver drowsiness to prevent car accidents and improve road safety. This study explores and proposes potential solutions to mitigate drowsiness-related accidents and enhance overall road security by monitoring the driver's eye, mouth, and head movements, which are key indicators of drowsiness. An algorithm has been developed to track these movements. The study involves analyzing CNN models and computer vision models for eye detection, yawn detection, and head movement, with the objective of identifying the most effective approach among the models. For building CNN models, we use the MRL Eye Dataset and YawDD datasets for eye and mouth tracking. The performance of the models developed using these datasets will be assessed based on various metrics such as accuracy and loss. We achieved an accuracy of 84.53% using the CNN approach for the MRL Eye Dataset, while for the YawDD Dataset, we obtained an accuracy of 96.42%. Despite these impressive results, CNN techniques encounter several challenges, including poor performance in low light conditions, varying outcomes for different ethnicities, and unreliability in dynamic environments. To address these challenges, we utilize Dlib and MediaPipe for tracking facial landmarks. By leveraging these techniques, we

can simultaneously monitor head movement, eye movements, and detect yawns. Between Dlib and MediaPipe, we prefer MediaPipe due to its capability to detect the 3D coordinates of key points.

Keywords: Drowsiness, CNN, CV approach, Dlib and Media pipe

1 Introduction

There are numerous factors that contribute to vehicle accidents, such as mechanical failures and failure to adhere to traffic signs and signals. However, drowsiness stands out as a primary cause of accidents and fatalities. Hence the development of a drowsiness alarm system for drivers presents a viable solution to mitigate accidents caused by drowsiness. Regarding driving safety, the threat of fatigued driving is a constant concern that poses a serious threat to commuters' safety around the globe. According to the Insurance Information Institute (III), in 2021, there were 1,310 incidents involving drowsy, asleep, fatigued, ill, or blacked out drivers. This highlights the significant risk posed by drowsy driving on roadways. Additionally, the National Highway Traffic Safety Administration (NHTSA) [7] provides essential insights into the prevalence and impact of drowsy driving Accidents. Figure.1 illustrating the statistical analysis of accidents results in drowsy statistics from year 2012 to 2021. From the statistical analysis it is clear that the rate of fatal crashes are increased around 12% from 2020 to 2021. Drowsiness detection can be categorized into vehicle based, behavioural

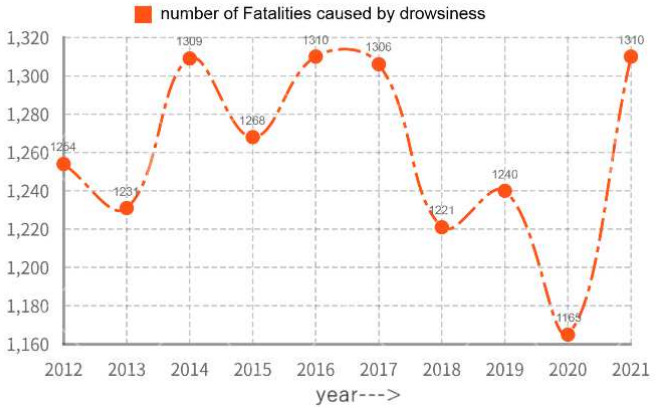


Fig. 1 The graph illustrates between the annual number of deaths caused by drowsy driving

Based and physiological Based [8]. Behavioural Based detection can be done based on movement of head, eye closure and yawn. Vehicle Based drowsiness detection it can be done by monitoring the vehicle's lane, speed etc. Physiological-based drowsiness detection depends on the correlation between EOG and ECG, which are physiological

signals [9]. All these methods or approaches have their own advantages and disadvantages. The main disadvantage of vehicle-based and physiological-based detection are, for physiological-based detection, the driver has to wear electrodes on his body [10], and to implement vehicle-based detection modifications required modifications should be made to the vehicle [11]. So we mainly focus on the behavior-based drowsiness detection system to build a detection system that is well enough to detect the drowsiness of the driver. There are two approaches that we are going to discuss, those are drowsiness detection using a convolutional neural network and other approaches on computer vision technology. To Detect Drowsiness of a person essential factors are eye tracking, yawn tracking, and head poss. In this paper we are try to investigate the feasibility of CNN approach in this domain and target to compare the performance of CNN approach with the proposed CV approach. In CV approach we aim to develop a detection system that is capable of effectively identifying driver drowsiness. The contributions of this paper are as follows:

1. Investigating the performance and feasibility of drowsiness detection using convolutional neural network and computer vision technology.
2. Propose an efficient drowsiness detection algorithm by leveraging the CV approach for eye tracking, yawn tracking, and head traits.

The rest of this paper is structured as follows: Section 2 discusses the related works closely aligned with the proposed work. Section 3 elaborates on the methodology, followed by the Results and discussions in Section 4, and the Conclusion with a focus on future works is summarized in Section 6.

2 Related Work

Driver drowsiness detection is a crucial area of research aimed at preventing accidents caused by sleepy driving. Various methods have been developed and evaluated, focusing on different metrics and employing a range of technologies from traditional image processing to advanced deep learning models. Fuletra et al. [2013] proposed a Convolutional Neural Network (CNN) model that integrates EEG and image processing for drowsiness detection. This model takes the driver's pulse and oxygen levels through EEG-based methods [1]. The study by Albadawi et al. [2022] presents and evaluates various sleepiness detection methods based on different metrics, including image-based, vehicle-based, biological-based, and hybrid-based measurements. Furthermore, an innovative vision-based driver attention monitoring system has been described by Alam L.et al. [2019]. This system categorizes drivers' attentional states into four groups: weariness, drowsiness, distracted driving, and attentive driving. The classification of attentional states is based on factors such as the frequency of yawns, gaze direction, and the percentage of eyelid closure over time (PERCLOS) [3]. M. Ramzan et al. [2019] reviewed existing supervised learning techniques used for drowsiness detection and also the pros and cons and comparative study of the diverse methods are discussed. Zhang et al. [2012] present a nonintrusive drowsiness recognition method by eye-tracking and image processing. They propose one robust eye detection to address the problems caused by changes in illumination and driver posture. The measures

they were considered are the percentage of eyelid closure, maximum closure duration, blink frequency, average opening level of the eyes, opening velocity of the eyes, and closing velocity of the eyes. These measures were combined using Fisher's linear discriminant functions by a stepwise method to reduce the correlations and then extract an independent index for the prediction [5].

Table 1 Comparison of the proposed system features and methodology used with that of various approaches existing

Research	Features Used	Methods
Alam [3]	Eyelid closure, Yawn frequency, Gaze direction	Computer Vision
Zhang [5]	Eyelid closure and velocity	Fisher's linear discriminant fun
Khunpisuth O [6]	Head tilting and eye Blinking	Raspberry Pi 3 module
Alameen S A [7]	Yawning frequency, Eye state, facial expressions	3D-CNN) and LSTM
Hashemi M [8]	Eye status classification	FD-NN,VGG16 and VGG19
Zhao Z [9]	Eyelid closure and Mouth opening degree	Convolutional network
Jahan I [10]	Eye state categorization	CNN models (VGG16, VGG19, and 4D)
Liu W [11]	Positioning mouth and eye	Convolutional neural networks (MTCNNs)
Reddy B [12]	Eyes, Mouth and Face	Deep neural networks model
Jabbar R [13]	Facial landmark	Deep neural networks model
Mehta S [14]	Facial landmarks, (EAR), (ECR)	Deep neural networks model
Knapik M [16]	Yawn detection	Neuro computing in thermal images
Abtahi S [17]	Yawning	Neural networks model
Murphy-Chutorian E [18]	Head Pose	Neural networks model
Fanelli G [19]	Head Pose	Random Regression Forests
Rocca F [20]	Head Pose	Neural networks model
Hammadi Y [21]	Head Pose	3DDFA_V2 algorithm
Ours	Eye aspect ratio, Mouth ratio and Head pose	CNN(YawDD&MRL Dataset)
Ours	Eye aspect ratio, Mouth ratio and Head pose	Computer Vision, Media Pipe

Khunpisuth et.al [2016] utilized Raspberry Pi Camera and Raspberry Pi 3 module, which can calculate the level of drowsiness in drivers. They used frequency of head tilting and blinking of the eyes to determine whether or not a driver felt drowsy. Alameen et.al [2023] propose a spatiotemporal model for monitoring drowsiness visual indicators from videos. They also utilized eye blinking and yawn for the prediction. Hashemi et.al [2020] introduced a potential network for eye status classification in which one of them is a Fully Designed Neural Network (FD-NN). Zhao et.al [2020] proposes a convolutional neural network, named EM-CNN, to detect the states of the eyes and mouth from the ROI images for make drowsiness. Jahan et.al [2023] propose a methodology that treats drowsiness detection as an object detection task, and from an incoming video stream of a driver they detect and localize open and closed eyes. MobileNet CNN architecture with Single Shot Multibox Detector (SSD) is used to perform this task. Liu et.al [2019] proposes an algorithm to positioning the mouth and eye with multi-task cascaded convolutional neural network and then extract the static and dynamic features from a partial facial image. By combining both static and dynamic features using a two-stream neural network they make the classification [11]. Reddy et.al [2017] introduced one minimized network structure based on facial landmark input to recognize whether driver is drowsy or not. Their proposed model

achieved an accuracy of 89.5% on 3-class classification and a speed of 14.9 frames per second (FPS) on Jetson TK1 [12]. Jabbar et.al [2018] propose a minimal network structure based on facial landmark key point detection to recognize whether the driver is drowsy. Their model can achieve an accuracy of more than 80% [13]. Mehta et.al [2019] propose a light-weight, real time driver's drowsiness detection system which is developed and implemented on Android application. Their system records the videos and detects driver's face in every frame by employing image processing techniques. They also detecting facial landmarks, computes Eye Aspect Ratio (EAR) and Eye Closure Ratio (ECR) to detect driver's drowsiness based on adaptive thresholding [14].

Although existing research has explored these indicators individually, there is a gap in the literature concerning their combined use. Integrating eye opening and closing, yawning, and head posture into a cohesive system could significantly improve the detection of driver drowsiness, enhancing road safety and reducing the risk of fatigue-related accidents. So in this research we focus on developing and testing multi-indicator systems to validate their effectiveness in real-world driving conditions.

3 Methodology

Driver Drowsiness Detection can be implemented with the help of Image Processing. For building models, we can adopt CNN or computer vision technology or both approaches can be implemented as depicted in Figure 2

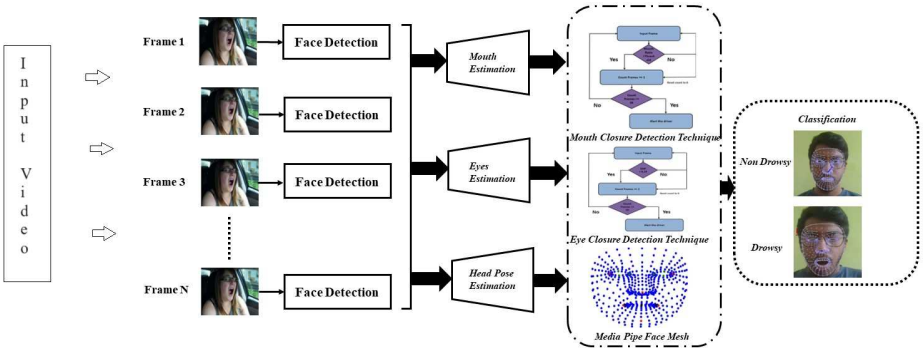


Fig. 2 The flow chart illustrating drowsiness detection system integrating eye opening and closing, yawning, and head posture into a cohesive system

3.1 CNN Approach:

In the CNN approach video frames are fed as input and the model detects drowsiness based on the trained data set. If the model predicts any drowsiness, then we will use

an alarm to warn the driver and prevent possible drowsiness accidents. The Data set that we are going to use to build a model is MRL eye data set. Total no. of images contained in the particular dataset involves 48000. Table 1 gives an overview of the details of MRL Eye Dataset.

The data set is divided into validation data and training data, then by using a sequential layer, the model is trained with 100 epochs. The workflow of sequential layers with input and output shapes is illustrated in Figure 3. We have a total of three convolution2d layers, three maxpooling2d layers, one Dropout, one Flatten, and Two Dense Layers. We start with an input shape of (32x32x3) and after the final dense layer, we end with an output shape of (2).

Table 2 Detailing the different input features of MRL eye dataset

Sl.No	Input Features	No.of Categories
1	Subject ID	xxx
2	Number of Images	48000
3	Gender	2 Categories, 1- Female, 0- Male
4	Glasses	2 Categories, 1- Yes, 0- No
5	Eye States	2 Categories, 1- Open, 0- Closed
6	Reflections	3 Categories, 2-High, 1- Low, 0- None
7	Lighting Condition or Image quality	2 Categories, 1- good, 0- bag

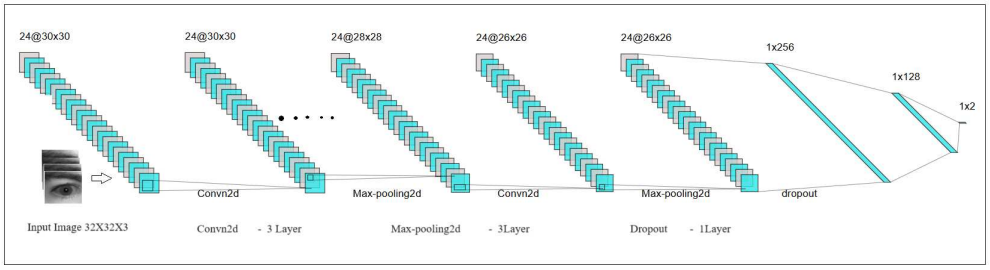


Fig. 3 The architecture schematics illustrating drowsiness detection using CNN method for eye detection

3.1.1 Detecting both eye movements and yawns simultaneously

Detecting both eye movements and yawns simultaneously is a critical aspect of identifying drowsiness. YawDD [18] is a dataset that can be utilized for tracking both eye and yawn movements. This dataset comprises images depicting opening and closing of eyes as well as images with and without yawns. In total the dataset contains 2900 images. Figure 4 illustrating YawDD dataset contains Eye and mouth features and Table 2 details the features of the dataset.



Fig. 4 Illustrating the features of YawDD dataset which represents both eye and mouth features [18]

Data augmentation is performed on the training dataset because of the limited no of sample in the training dataset. The sequential layers as depicted in Figure 5, illustrate the workflow with input and output feature dimensions.

Table 3 Detailing the different input features of YawDD dataset used for detecting both eye movements and yawns simultaneously

Sl.No	Input Features	No.of Categories
1	Subject ID	xxx
2	Number of Images	2900
3	Gender	2 Categories, 1- Female, 0- Male
4	Glasses	2 Categories, 1- Yes, 0- No
5	Eye States	2 Categories, 1- Open, 0- Closed
6	Mouth States	3 Categories, 1- no talking, 2- talking or singing, 3- yawning
7	Reflections	3 Categories, 2-High, 1- Low, 0- None

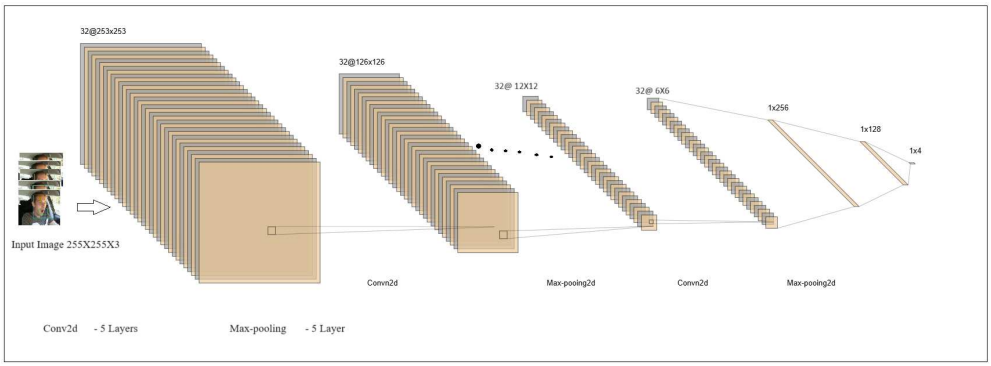


Fig. 5 Illustrating the architecture schematics for Eye and Yawn Detection

3.2 CV Approach:

3.2.1 Method for Eye Tracking

The PERcentage of eyelid CLOSure (PERCLOS) method in computer vision is used to calculate the percentage of eyelid closure. This enables us to determine whether the eye is open or closed [9]. Eye Aspect Ratio (EAR) is the parameter that is used to find whether an eye is closed or not [5]. It is calculated based on the distances between facial landmarks around the eye. The formula to calculate the EAR for an eye is as given in equation (1) [5]. An EAR value of '0.25' indicates opening state of the eye . while an EAR below this threshold signifies a closed eye. The algorithm counts for each frame with an EAR value less than .25 and when this count exceeds 20 the system alerts the driver with an alarm. Figure 7 represents the flow chart illustrating the eye closure detection. In order to accomplish this result it is necessary to locate both eyes exactly. This can be achieved by utilizing Dlib [24] and Mediapipe [23] as illustrated in Figure 6.



Fig. 6 Illustrating the terms used to formulate the expression for Eye Aspect Ratio [5]

$$\text{EAR} = \frac{\left\|\tilde{P}2 - P6\right\| + \left\|\tilde{P}3 - P5\right\|}{2 \cdot \left\|\tilde{P}1 - P4\right\|} \tag{1}$$

where P1, P2, P3, P4, P5 and P6 are the coordinates of the eye landmarks as illustrated in Figure 6. and

$$\|P_i - P_j\|$$

represents the Euclidean distance between the points Pi and Pj.

3.2.2 Method for Mouth Tracking

The parameter known as mouth ration is used to find whether a mouth is closed or not. The Mouth ratio is obtained by dividing the height of the mouth by its width [3]. The Dlib and Mediapipe libraries provide facial landmark features, but we only focus on the mouth features from the facial landmarks. In order to detect yawn we

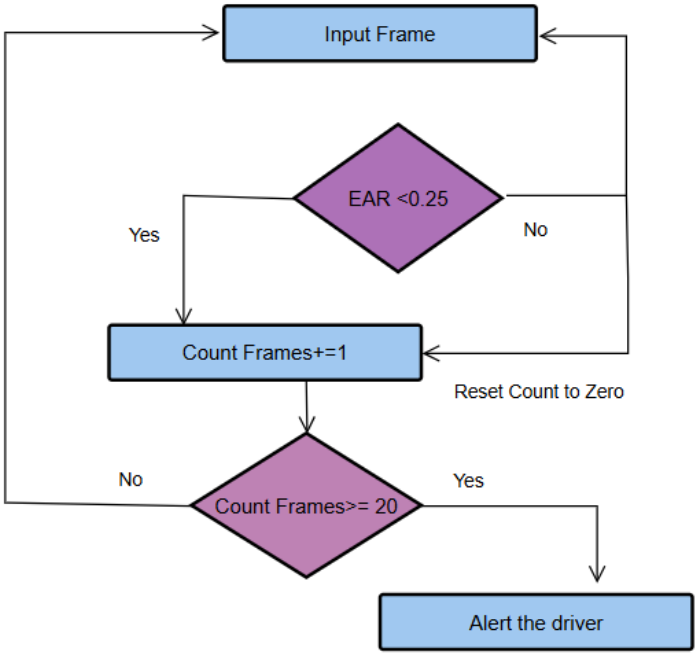


Fig. 7 Illustrating the Flow chart which represents the eye closure detection

consider the top and bottom features of the lips to determine the separation between upper and lower lips or height of the mouth, and the left and right points of the lips to determine its width. Subsequently the ratio of the height to the width is calculated using equation (2) and the threshold of 0.3 is used to classify whether mouth is open or close [4]. If the mouth ratio is less than 0.3, we classify it as closed mouth. And if the mouth ratio is more than 0.3, we classify it as open mouth. For each time the mouth ration detected to be greater than the threshold of 0.3 the counter will update the frame count and if the frame count exceeds 36 the system will alert the driver with an alarm. Figure 8 represents the flow chart illustrating the yawn detection process.

$$\mathrm{MR} = \frac{HM}{WM} \tag{2}$$

where MR, HM and WM represents mouth ration, Hight of the mouth and width of the mouth respectively.

3.2.3 Method for Head Pose Estimation

In order to utilize Head Pose Estimation, it is essential to monitor the movement of the head [19]. To accomplish this, Mediapipe can be employed, as it enables the detection of facial landmarks and expressions. Our objective is to estimate the position of the head by utilizing six points from the face mesh, as depicted in Figure 9. By employing

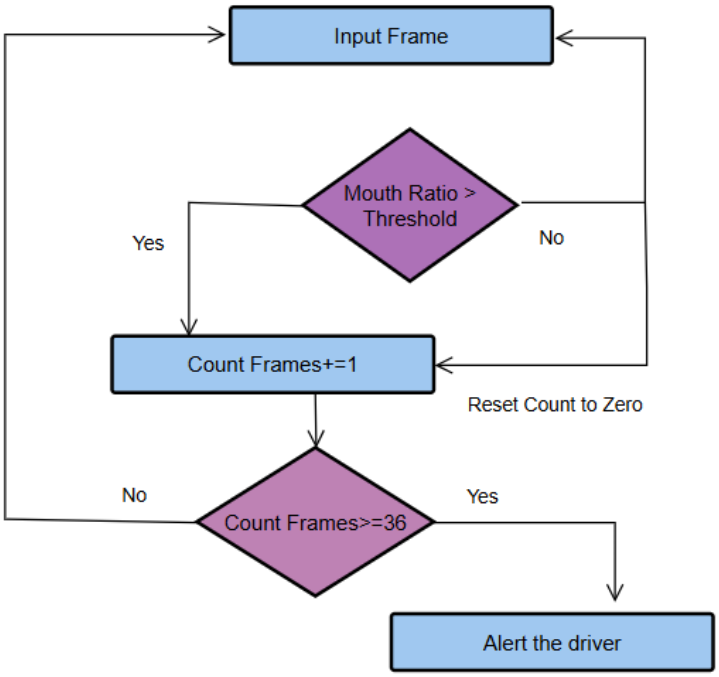


Fig. 8 Illustrating the Flow chart which represents the Yawn detection

the SolvePnP (Perspective n Point) method, we can estimate the camera calibration based on the given points and obtain the rotation values as roll, pitch, and yaw [21].

The rotation angle is then utilized to estimate the coordinates (x, y, z) of the driver's gaze. If the value of y falls within the range of -10 to 10, it indicates that the driver is looking straight ahead; otherwise, they are distracted. If the value of X and Y is detected to be out of the range between -10 to 10 then the Head pose counter will increment its count value. When ever the Head pose count value exceeds a predefined threshold of 25, the system alert the driver with an alarm. Other inferences regarding the values of x and Y are as follows. if x is greater than 10, it implies the driver is looking upwards; if x is less than -10, it indicates the driver is looking downwards; if y is greater than 10, it signifies the driver is looking to the right; and if y is less than -10, it suggests the driver is looking to the left. Table 3 outlines the thresholds used to make these predictions.

Figure 10 illustrates the flow chart which represents the driver drowsiness detection in the computer vision approach through the concurrent monitoring of eye movement, yawn detection, and head gestures. Eye movement analysis helps in identifying if the driver's eyes are closed for extended periods, which is a sign of drowsiness. Yawn detection involves recognizing the driver's mouth opening frequently and widely, which is also another sign of fatigue. Head gesture analysis detects unusual head movements, such as nodding or tilting, which can also indicate drowsiness. By simultaneously tracking all these indicators, the system can provide a more accurate and reliable

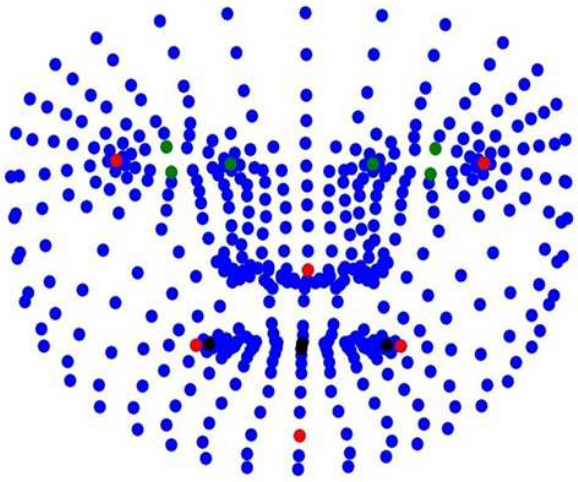


Fig. 9 Illustrating Media Pipe Face Mesh with 6 red points for Head Pose Estimation

Table 4 Different thresholds considered for X and Y to make the predictions on head poss

Sl.No	Range of X	Range of Y	Head Poss
1	_	$-10 \leq Y \leq 10$	Forward
2	$X > 10$	_	Up
3	$X < -10$	_	Down
4	_	$Y > 10$	Right
5	_	$Y < -10$	Left

assessment of the driver's state of alertness. This multimodal approach helps in reducing false positives and improving the reliability of drowsiness detection, ultimately enhancing road safety.

The complete algorithm for facial landmark detection and the drowsy driver alert system is detailed in Algorithm (1). This algorithm combines facial landmark detection, head pose estimation, and eye and mouth aspect ratio calculations to detect drowsiness in drivers.

4 Result and Discussion

4.1 Driver alertness system in CNN approach:

The dataset has been divided into three distinct subsets such as training, validation, and test. Initially, the model undergoes training utilizing the data from both the training and validation subsets to optimize its parameters and improve its capability to generalize. Following this training phase, the test dataset is employed to assess the model's performance, indicating how well the model is likely to perform on unseen data.

Algorithm 1 Facial Landmark Detection and Alert System

Require: cv2, time, numpy, mediapipe, pygame.mixer libraries

Initialize Libraries and Alert Sound
Import necessary libraries: cv2, time, numpy, mediapipe, pygame.mixer
Initialize the alert sound using `mixer.init()` and load the alert sound file
Initialize Mediapipe FaceMesh
Initialize Mediapipe FaceMesh for facial landmark detection and tracking
Set minimum detection and tracking confidence thresholds
Initialize Drawing Utilities
Initialize drawing utilities for drawing facial landmarks and lines
Initialize Eye and Yawn Thresholds and Counters
Set thresholds for eye aspect ratio (EAR) and yawn detection
Set the number of consecutive frames to consider for eye closure and yawn detection
Reset counters to track consecutive frames of eye closure and yawning.
Define Facial Feature Landmarks
Define indices for facial landmarks corresponding to head pose, left eye, right eye, and mouth
Start Video Capture
Open the default camera for capturing video frames
Main Loop for Video Processing
while video stream is open **do**
    Read frames from the video stream
    Flip the frames horizontally for selfie-view display
    Convert the frames to RGB for processing with Mediapipe FaceMesh
    Process the frames with Mediapipe FaceMesh to track facial landmarks
    Convert the frames back to BGR color space for display
    Extract facial landmarks for each detected face in the frame
    Head Pose Estimation
    Extract relevant facial landmarks for head pose estimation
    Calculate head pose angles using `solvePnP` and `Rodrigues` rotation
    Eye Aspect Ratio (EAR) Calculation
    Extract landmarks for the left and right eyes
    Calculate the eye aspect ratio (EAR) for each eye
    Determine eye closure based on the EAR values
    Yawn Detection
    Extract landmarks for the mouth
    Calculate the mouth aspect ratio for yawn detection
    Determine yawn based on the mouth aspect ratio
    Display Results and Alerts
    Display head direction, eye closure status and yawn detection status
    Draw lines to visualize head the pose direction
    Draw facial landmarks and lines on the frame
    Display the frame with overlays and alerts
end while
Handle User Input
Wait for the 'esc' key to be pressed to exit the program

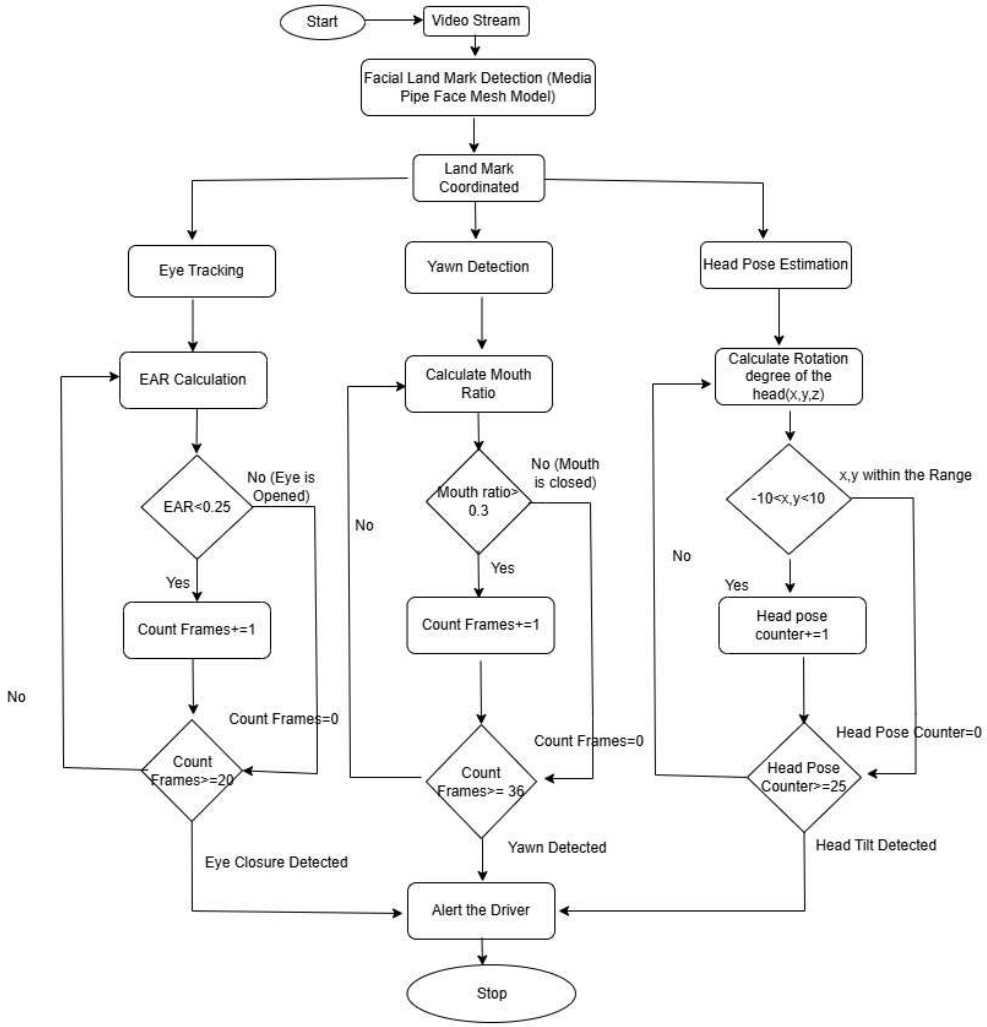


Fig. 10 Illustrating the Flow chart which represents the driver drowsiness detection in CV approach

Figure 11(a) and Figure 11(b) display the performance metrics—specifically the accuracy and loss—of the model trained on the yawdd dataset. These figures help visualize how effectively the model predicts the correct outcomes and how its error metrics decrease over time during training.

Similarly, Figure 11(c) and Figure 11(d) focus on the model trained using the MRL dataset. These figures also depict the accuracy and loss during the training and validation phases, offering insights into the model's learning efficiency and its capability to handle data from this specific dataset. The graphical representations serve as a useful tool for evaluating the model's performance dynamics throughout the training period.

From various drowsiness detection techniques, we find that various CNN techniques face multiple issues such as performing poorly in low light environments, different results for different ethnicities and being unreliable in moving environments. The dataset 'YawDD' is limited to 2900 different images. Coming to Computer vision technique, the dlib frontal features include 68 facial landmarks on a particular face. But it performs poorly for head pose estimation with its unstable landmarks while turning the head [24]. So we use the library 'MediaPipe' [23], which includes a 4783 D facial landmarks. The face mesh imposed on the driver's face helps us get the eye and mouth features for eye closure and yawn detection. And the 6 facial points for head pose estimation are stable and available even if the head is turned away from the camera. And Mediapipe performs well in all lighting conditions and for all ethnicities.

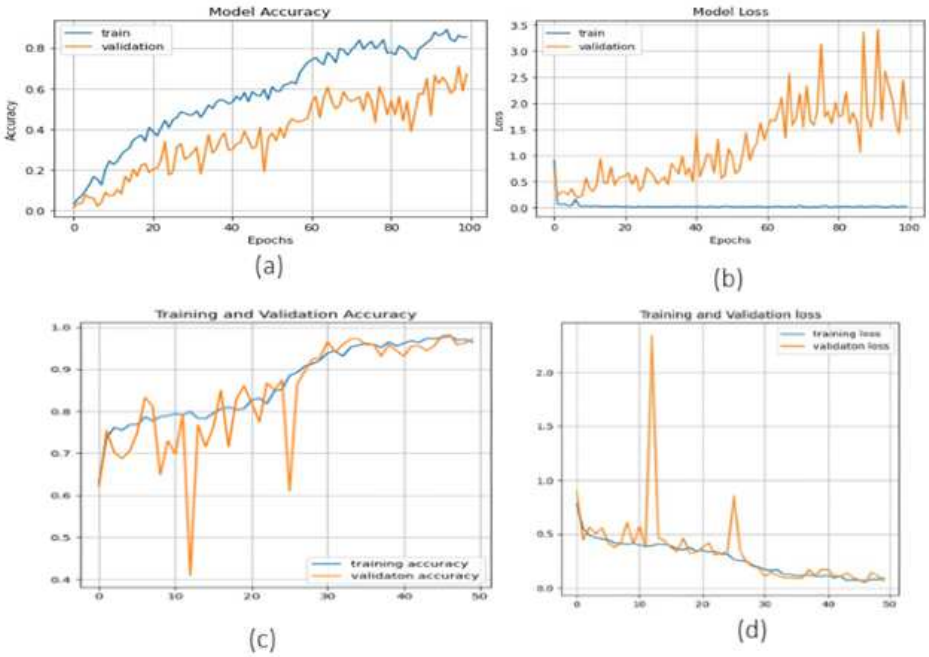


Fig. 11 (a) Illustrate the accuracy of the model constructed on the yawdd dataset (b) illustrate the loss of the model constructed on the yawdd dataset,(c)illustrate the accuracy of the model using the MRL dataset during training and validation, (d) illustrate the loss of the model using the MRL dataset during training and validation

4.2 Driver alertness system in CV approach:

Using computer vision technology generally yields superior results compared to conventional Convolutional Neural Network (CNN) approaches for certain applications because each dataset has its own unique challenges and limitations. Computer vision

techniques excel particularly in the task of detecting facial landmarks, a critical operation in monitoring driver alertness. For this purpose, we employ the Mediapipe framework, which is specifically designed to handle complex facial recognition tasks efficiently.

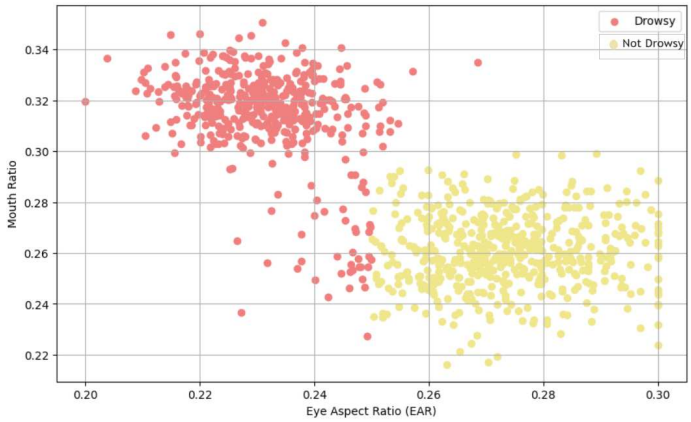


Fig. 12 Scatter plot illustrating drowsy detection for EAR value less than 0.25 and Mouth ratio greater that 0.3

While driving, the primary responsibility of the driver is to maintain attention on the road. To assist in avoiding instances of drowsiness or distraction, it is necessary to monitor various physiological and behavioral indicators through real-time computer vision analysis. These indicators include yawning, head pose, and eye movements. Each frame of the video captured from the driver's environment is analyzed to calculate three key metrics: the Eye Aspect Ratio (EAR) to assess eye closures, the Yawn Ratio to detect yawning, and Head Pose Estimation to determine the orientation of the driver's head.

Consider the scenario depicted in Figure 12 (b) and 12 (c), where it is evident that the driver is distracted and not focusing on the road. In such cases, the system might designed to trigger an alert. Similarly in Figure 12 (d) the system identified Yawning by the driver. If its frequency exceeds the predefined threshold, an alert will be enabled. This alert functions as a critical wake-up call to the driver, urging them to either re-engage actively with the task of driving or to take a break if necessary. The system's ability to detect both distraction and signs of fatigue in real-time and respond with appropriate alerts plays a vital role in enhancing road safety. By addressing these key indicators of driver inattention and fatigue, the system helps maintain the alertness of the driver, thereby significantly minimizing the likelihood of accidents caused by drowsiness or inattention.

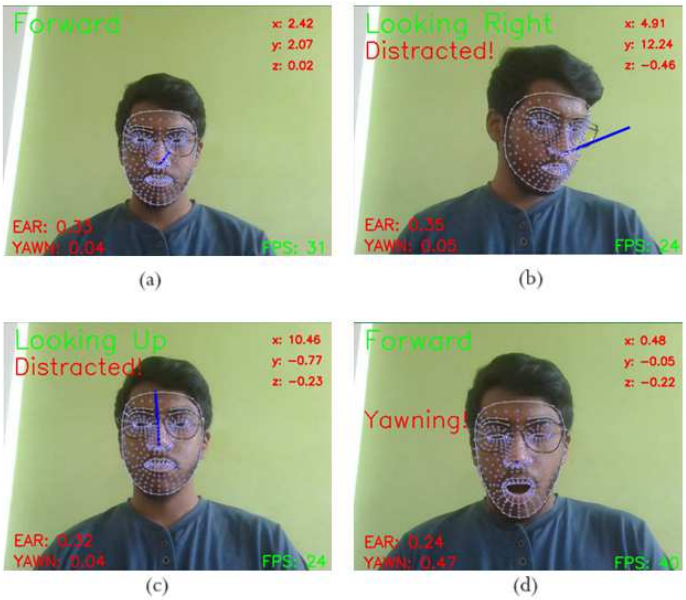


Fig. 13 (a) Person looking forward (b) Person distracted and looking right(c) Person distracted and looking up (d) Person Yawning

5 Conclusion

Driver drowsiness detection systems are critical safety technologies proposed to improve road safety by detecting signs of fatigue or distraction in drivers. These sophisticated systems employ a mix of sensors, cameras, and advanced algorithms to closely monitor the driver's behavior and initiate timely alerts to mitigate the risk of accidents.

In recent evaluations using various approaches and datasets, the results have been promising. For instance, employing a Convolutional Neural Network (CNN) approach on the MRL Eye Dataset yielded an accuracy of 84.53%, indicating a high level of effectiveness in eye tracking for signs of drowsiness. Similarly, when utilizing a CNN for the YAWDD (Yawning Detection and Drowsiness) dataset, the system achieved an impressive accuracy of 96.42%, successfully identifying yawn occurrences, which are strong indicators of fatigue.

In addition to these methods, facial landmark tracking technologies like Dlib and Mediapipe have been crucial. These tools are adept at identifying facial landmarks, which are then used to monitor eye movements, head positioning, and yawn detection. Of these two, Mediapipe has shown particular promise due to its ability to capture three-dimensional coordinates of facial key points. This feature greatly enhances the system's ability to accurately track subtle movements and changes in the driver's face, providing a more nuanced understanding of their state.

By continuously analyzing various physiological and behavioral indicators, these systems can alert drivers at the first signs of drowsiness or distraction, urging them to take necessary actions—such as taking a break or refocusing on the road—which substantially increases overall road safety.

Supplementary information.

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Authors' contributions. AKR: Conceptualization, Methodology, Software, Writing-original draft, Writing-Review & Editing. IAK: Methodology, Analysis, Investigation, Software, Writing-original draft, Writing-Review & Editing. NCT: Methodology, Resources, Software, Validation, Writing-Review & Editing. LLT: Project administration, Supervision, Validation, Writing-Review & Editing. AMA: Writing-Review & Editing. AK: Resources, Writing-Review & Editing. SK: Resources, Writing-Review & Editing.

References

[1] Fuletra J D., & Bosamiya D (2013) A survey on drivers drowsiness detection techniques. International Journal on Recent and Innovation Trends in Computing and Communication, 1(11):816-819

[2] Albadawi Yaman, Maen Takruri, and Mohammed Awad (2022) A Review of recent developments in driver drowsiness detection systems. Sensors 22(5): 2055-2069

[3] Alam L, Hoque M M (2019) Vision-based driver's attention monitoring system for smart vehicles. In: Intelligent Computing & Optimization, Springer, Cham, pp 196-209

[4] Ramzan M, Khan H U, AwanS M, Ismail A, Ilyas M and Mahmood A (2019) A survey on state-of-the-Art drowsiness detection techniques. IEEE Access 7: 61904-61919

[5] Zhang W, Cheng B, Lin, Y (2012) Driver drowsiness recognition based on computer vision technology. Tsinghua Science and Technology 17(3):354-362

[6] Khunpisuth O, Chotchinasri T, Koschakosai V, Hnoohom N (2016) Driver drowsiness detection using eye-closeness detection. In: 12th International Conference on Signal-Image Technology & Internet-Based Systems (SITIS), IEEE, Italy, pp 661-668

[7] Alameen S A, Alhothali, A M (2023) A lightweight driver drowsiness detection system using 3DCNN with LSTM. Computer Systems Science & Engineering, 44(1): 895-912. https://doi.org/10.32604/csse.2023.024643

[8] Hashemi M, Mirrashid A, Beheshti Shirazi A (2020) Driver safety development: real-time driver drowsiness detection system based on convolutional neural network. SN Computer Science, 1(5): 289. https://doi.org/10.1007/s42979-020-00306-9

[9] Zhao Z, Zhou N, Zhang L, Yan H, Xu Y, Zhang Z (2020) Driver fatigue detection based on convolutional neural networks using EM-CNN. Computational intelligence and neuroscience, 1-11. https://doi.org/10.1155/2020/7251280

[10] Jahan I, Uddin K A, Murad S A, Miah M S U, Khan T Z, Masud M, Bairagi A K (2023) 4D: 4D: A real-time driver drowsiness detector using deep learning. Electronics, 12(1): 235. https://doi.org/10.3390/electronics12010235

[11] Liu W, Qian J, Yao Z, Jiao X, Pan J (2019) Convolutional two-stream network using multi-facial feature fusion for driver fatigue detection. Future Internet, 11(5):115. https://doi.org/10.3390/fi11050115

[12] Reddy B, Kim Y H, Yun S, Seo C, Jang J (2017) Real-time driver drowsiness detection for embedded system using model compression of deep neural networks. In: conference on computer vision and pattern recognition workshops, IEEE USA, pp 121-128. http://dx.doi.org/10.1109/CVPRW.2017.59

[13] Jabbar R, Al-Khalifa K, Kharbeche M, Alhajyaseen W, Jafari M, Jiang S (2018) Real-time driver drowsiness detection for android application using deep neural networks techniques. In: 9th International Conference on Ambient Systems, Networks, and Technologies (ANT), 130, Elsevier Portugal, pp 400-407, https://doi.org/10.1016/j.procs.2018.04.060

[14] Mehta S, Dadhich S, Gumber S, Jadhav Bhatt A (2019) Real-time driver drowsiness detection system using eye aspect ratio and eye closure ratio. In: Proceedings of international conference on sustainable computing in science, technology and management (SUSCOM), India. Pp 1333-1339. https://dx.doi.org/10.2139/ssrn.3356401

[15] Lee A (2015) Comparing deep neural networks and traditional vision algorithms in mobile robotics. Swarthmore University, 40

[16] Knapik M, Cyganek B (2019) Driver's fatigue recognition based on yawn detection in thermal images. Neurocomputing, 338: 274-292

[17] Abtahi S, Omidyeganeh M, Shirmohammadi S, Hariri B. (2014) YawDD: A yawning detection dataset. In: proceedings of 5th ACM multimedia systems conference, ACM Singapore, pp 24-28

[18] Murphy-Chutorian E,Trivedi M M (2008) Head pose estimation in computer vision: A survey. IEEE transactions on pattern analysis and machine intelligence 31(4): 607-626

[19] Fanelli G, Gall J, Van Gool L (2011) Real time head pose estimation with random regression forests. In: 28th Conference of Computer vision and pattern recognition (CVPR), IEEE USA, pp 617-624

[20] Rocca F, Mancas M, Gosselin B (2014) Head , pose estimation by perspective-npoint solution based on 2d markerless face tracking. In: 6th International Conference of Intelligent Technologies for Interactive Entertainment (INTETAIN) SPRINGER USA. pp. 67-76

[21] Hammadi Y, Grondin F, Ferland F, Lebel K (2022) Evaluation of various state of the art head pose estimation algorithms for clinical scenarios. Sensors, 22(18): 6850

[22] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, EshaUboweja, Michael Hays, Fan Zhang, et al. (2019) Mediapipe: A framework for building perception pipelines. arXiv preprint arXiv:1906.08172

[23] Davis E King (2009) Dlib-ml: A machine learning toolkit. The Journal of Machine Learning Research, 10: 1755–1758